German University in Cairo (GUC)
Faculty of Information Engineering & Technology
Department of Communications — Communication Theory

Assignment 1: Uniform vs. K-means Quantization

Simulation, Quantitative Evaluation, and Comparative Analysis

Group Member Name	University ID
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Sama Ismael Ahel	64-19880

1. Parameter Setup & Signal Generation

The assignment requires generating a random vector of N = 10,000 samples uniformly distributed in [-a, a], where a is the largest single digit among the group members' university IDs:
• Mazen Altelbany (64-12371) → Digits: {1, 2, 3, 4, 6, 7}
• Suhad Eyhab Rasheed (64-31506) → Digits: {0, 1, 3, 4, 5, 6}
• Sama Ismael Ahel (64-19880) → Digits: {0, 1, 4, 6, 8, **9**}

The highest individual digit across all members is **9**. Therefore: **a = 9**, giving a dynamic range of **[-9, 9]**.
Theoretical signal power is $P_{signal} = a^2 / 3 = 9^2 / 3 =$ **27.0000**. Our simulated signal power from 10,000 samples is **27.05**.

2. Task 1: First 6 Elements of Original & Quantized Vectors

The table below presents the first six elements of x alongside the quantized vectors for L = 2, 4, and 8:

Sample #	Original x	Uniform (L=2)	K-means (L=2)	Uniform (L=4)	Uniform (L=8)	K-means (L=8)
1	-2.2583	-4.5000	-4.5481	-2.2500	-3.3750	-1.3646
2	8.1129	4.5000	4.4005	6.7500	7.8750	7.8718
3	4.1759	4.5000	4.4005	2.2500	3.3750	3.2301
4	1.7759	4.5000	4.4005	2.2500	1.1250	0.9032
5	-6.1917	-4.5000	-4.5481	-6.7500	-5.6250	-5.7874
6	-6.1921	-4.5000	-4.5481	-6.7500	-5.6250	-5.7874

3. Task 2: Mean Absolute Quantization Error (MAE)

Levels (L)	Step Size (Δ)	Uniform MAE	K-means MAE
L = 2	9.00	2.2606	2.2600
L = 4	4.50	1.1220	1.1217
L = 8	2.25	0.5580	0.5575

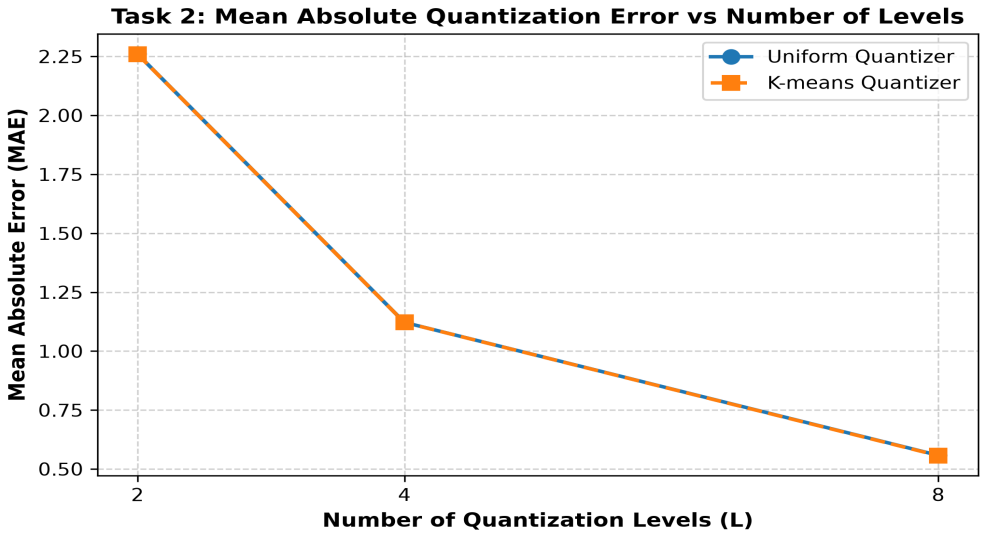


Figure 1: Mean Absolute Quantization Error vs. Number of Levels.

Performance Comments (Task 2):

• **Halving of Error:** When L doubles, the step size Δ is halved. Since error is uniform in $[-\Delta/2, \Delta/2]$, the mean absolute error is approximately $\Delta/4$. MAE reduces by ~50% at each step.

• **Equivalence:** Both quantizers perform identically because the uniform PDF makes equal intervals optimal.

4. Task 3: Quantization Error Variance

Levels (L)	Simulated Uniform	Simulated K-means	Theoretical ($\Delta^2/12$)
L = 2	6.7857	6.7842	6.7500
L = 4	1.6766	1.6754	1.6875
L = 8	0.4179	0.4191	0.4219

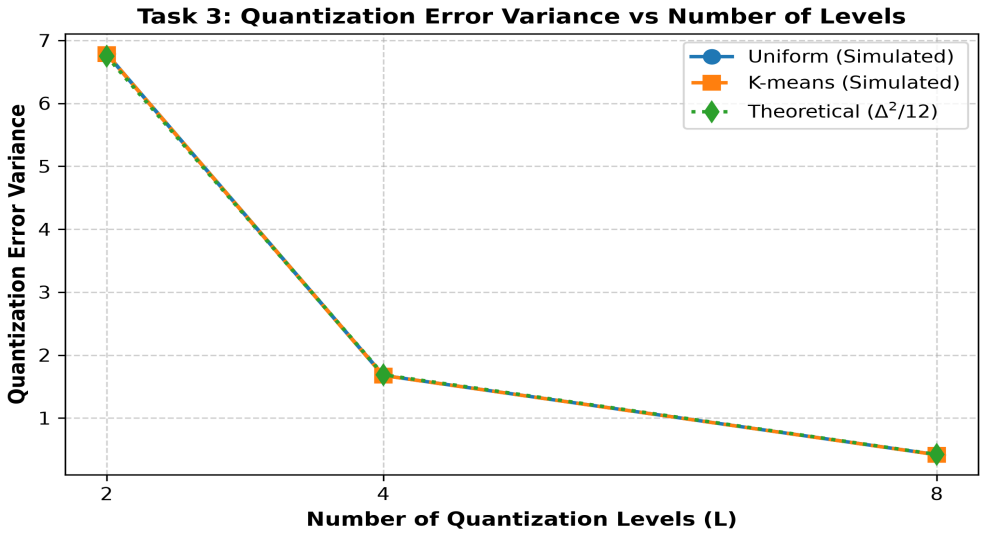


Figure 2: Quantization Error Variance vs. Number of Levels.

Performance Comments (Task 3):

• **Quadratic Drop:** Doubling L divides Δ by 2, causing the variance (proportional to Δ^2) to decrease by a factor of 4 (75% drop).

• **Close Fit:** The simulated sample variances for both uniform and K-means quantizers match the theoretical formula ($\Delta^2/12$) very tightly.

5. Task 4: Signal-to-Quantization-Noise Ratio (SQNR)

Levels (L)	Bits (R)	Simulated Uniform	Simulated K-means	Theoretical SQNR
L = 2	1 bit	5.97 dB	5.97 dB	6.02 dB
L = 4	2 bits	12.04 dB	12.04 dB	12.04 dB
L = 8	3 bits	18.07 dB	18.06 dB	18.06 dB

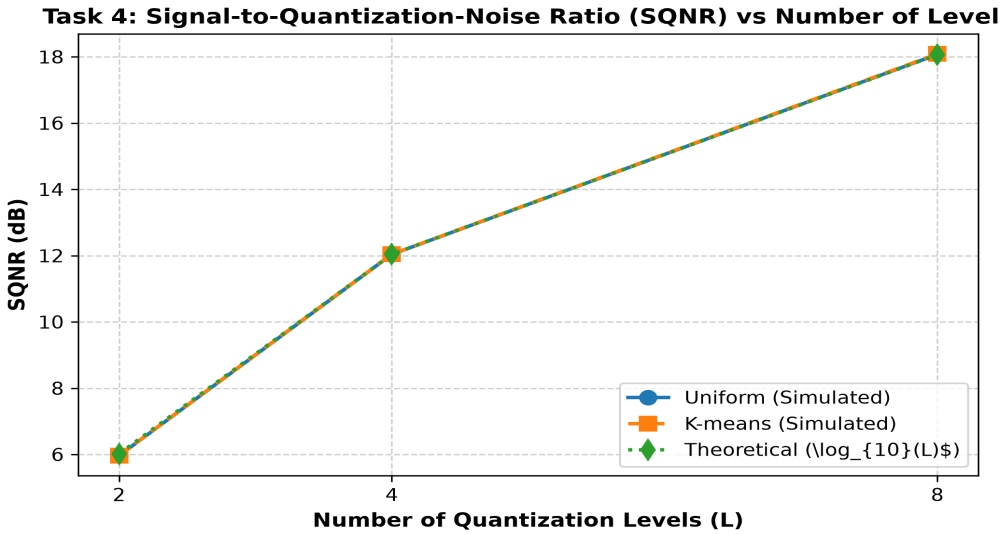


Figure 3: Signal-to-Quantization-Noise Ratio (SQNR) vs. Number of Levels.

Performance Comments (Task 4):

- **The 6.02 dB Rule:** Each extra bit of quantization resolution doubles the number of levels and increases SQNR by ~6.02 dB.
- **Matching Curves:** Both quantizers match the theoretical SQNR line (20·log10(L)) across all tested levels.

6. Discussion: Comparison of the Two Approaches

How Quantization Levels are Determined:

• **Uniform Quantizer:** Determined analytically prior to data observation. The full dynamic range [$-a$, a] is divided into L equal sub-intervals with step size $\Delta = 2a/L$. The thresholds are $-a + k\Delta$, and levels are the arithmetic midpoints.

• **K-means Quantizer:** Determined iteratively from the data points (Lloyd-Max algorithm). It starts with initial centroid positions and iterates two steps until convergence: (1) Nearest neighbor assignment (samples are partitioned into Voronoi cells), and (2) Centroid update (each centroid is recalculated as the mean of points in its cell).

Why They Yield Identical Performance on this Signal:

The input signal is uniformly distributed over [-9, 9]. When the probability density function is flat, the optimal Lloyd-Max centroids are mathematically equidistant midpoints. Thus, K-means naturally converges to the exact uniform levels.

Behavior on Non-Uniform Signals (e.g. Gaussian or Speech):

If the input were Gaussian or Laplacian speech, most samples would cluster near zero. A uniform quantizer would waste levels on rare extreme values, causing high distortion in the dense center. In contrast, K-means would place many closely spaced centroids near zero and few at the tails. This data-driven spacing minimizes mean squared error and achieves vastly superior SQNR over uniform quantization on non-uniform data.

7. Appendix: MATLAB Source Code (Mazen_Altelbany.m)

```
%% Communication Theory - Assignment 1
% Simulation and Comparison of Uniform and K-means Quantization
% Group Members:
%   1. Mazen Altelbany (64-12371)
%   2. Suhad Eyhab Rasheed (64-31506)
%   3. Sama Ismael Ahel (64-19880)
% Largest digit across IDs: a = 9 (from ID 64-19880) -> Range [-9, 9]

clear;
clc;
close all;

%% 1. Simulation Setup & Input Signal Generation
% Range parameter determined by maximum digit across all IDs
a = 9;

% Number of samples
N = 10000;

% Set random seed for repeatable results across runs
rng(42);

% Generate uniformly distributed input signal between [-a, a]
% Formula: x = -a + 2*a*rand(N, 1)
x = -a + 2 * a * rand(N, 1);

% Number of quantization levels to test (adaptable to any levels vector)
levels = [2, 4, 8];
numCases = length(levels);

% Preallocate metric arrays
mae_uniform  = zeros(1, numCases);
mae_kmeans   = zeros(1, numCases);

var_uniform  = zeros(1, numCases);
var_kmeans   = zeros(1, numCases);
var_theory   = zeros(1, numCases);

sqnr_uniform = zeros(1, numCases);
sqnr_kmeans  = zeros(1, numCases);
sqnr_theory  = zeros(1, numCases);

% Signal power: simulated and theoretical (for uniform distribution P_sig = a^2 / 3)
P_signal_sim    = mean(x.^2);
P_signal_theory = (a^2) / 3;

fprintf('================================================================================\n');
fprintf('           COMMUNICATION THEORY - ASSIGNMENT 1 SIMULATION              \n');
fprintf('  Signal Range: [-%d, %d] | Number of Samples: %d | Signal Power: %.4f \n', ...
        a, a, N, P_signal_sim);
fprintf('================================================================================\n\n');

%% 2. Loop Through Quantization Levels (L = 2, 4, 8)
for i = 1:numCases
    L = levels(i);

    fprintf('--------------------------------------------------------------------------------\n');
    fprintf('                            TESTING L = %d LEVELS                          \n', L);
    fprintf('--------------------------------------------------------------------------------\n');

    %% --- Part A: Uniform Quantizer ---
    % Step size (Delta)
    delta = (2 * a) / L;

    % Mid-point reconstruction levels for each interval
    uniform_levels = -a + ((1:L) - 0.5) * delta;

    % Map each sample to its corresponding interval index (1 to L)
    interval_idx = floor((x - (-a)) / delta) + 1;
    interval_idx(interval_idx > L) = L;
    interval_idx(interval_idx < 1) = 1;

    % Uniform quantized vector (guaranteed column vector)
    x_q_uniform = uniform_levels(interval_idx);
    x_q_uniform = x_q_uniform(:);

    %% --- Part B: Machine-Learning K-means Quantizer ---
    % Using MATLAB built-in kmeans function with multiple replicates
    % This lets K-means cluster the data independently from scratch
```

```
    [kmeans_idx, kmeans_centers] = kmeans(x, L, 'Replicates', 5, 'MaxIter', 100);

    % K-means quantized vector
    x_q_kmeans = kmeans_centers(kmeans_idx);
    x_q_kmeans = x_q_kmeans(:);

    %% --- Task 1: Display First 6 Samples ---
    fprintf('\n[Task 1] First 6 Elements Comparison (L = %d):\n', L);
    sample_table = table((1:6)', x(1:6), x_q_uniform(1:6), x_q_kmeans(1:6), ...
        'VariableNames', {'Sample_Index', 'Original_x', 'Uniform_x_q', 'Kmeans_x_q'});
    disp(sample_table);

    %% --- Task 2: Mean Absolute Error (MAE) ---
    e_uniform = x - x_q_uniform;
    e_kmeans  = x - x_q_kmeans;

    mae_uniform(i) = mean(abs(e_uniform));
    mae_kmeans(i)  = mean(abs(e_kmeans));

    fprintf('[Task 2] Mean Absolute Error (MAE):\n');
    fprintf('         Uniform Quantizer : %.4f\n', mae_uniform(i));
    fprintf('         K-means Quantizer : %.4f\n', mae_kmeans(i));

    %% --- Task 3: Quantization Error Variance ---
    var_uniform(i) = var(e_uniform);
    var_kmeans(i)  = var(e_kmeans);
    var_theory(i)  = (delta^2) / 12;

    fprintf('[Task 3] Quantization Error Variance:\n');
    fprintf('         Uniform Quantizer (Simulated)   : %.4f\n', var_uniform(i));
    fprintf('         K-means Quantizer (Simulated)   : %.4f\n', var_kmeans(i));
    fprintf('         Theoretical Variance (Delta^2/12): %.4f\n', var_theory(i));

    %% --- Task 4: Signal-to-Quantization-Noise Ratio (SQNR) ---
    P_noise_u = mean(e_uniform.^2);
    P_noise_k = mean(e_kmeans.^2);

    sqnr_uniform(i) = 10 * log10(P_signal_sim / P_noise_u);
    sqnr_kmeans(i)  = 10 * log10(P_signal_sim / P_noise_k);
    sqnr_theory(i)  = 10 * log10(P_signal_theory / var_theory(i));

    fprintf('[Task 4] Signal-to-Quantization-Noise Ratio (SQNR):\n');
    fprintf('         Uniform Quantizer (Simulated) : %.2f dB\n', sqnr_uniform(i));
    fprintf('         K-means Quantizer (Simulated) : %.2f dB\n', sqnr_kmeans(i));
    fprintf('         Theoretical SQNR              : %.2f dB\n\n', sqnr_theory(i));
end

%% 3. Comments and Performance Discussion Output to Console
fprintf('================================================================================\n');
fprintf('                            SUMMARY & COMMENTS                                  \n');
fprintf('================================================================================\n');
fprintf('1. MAE Performance (Task 2):\n');
fprintf('   - As the number of levels L doubles (2 -> 4 -> 8), step size Delta is halved.\n');
fprintf('   - Consequently, MAE drops by approximately 50%% at each doubling.\n');
fprintf('   - Both quantizers produce almost identical MAE across all tested levels.\n\n');

fprintf('2. Variance Performance (Task 3):\n');
fprintf('   - Error variance drops by a factor of 4 (approx 75%% decrease) when L doubles,\n');
fprintf('     consistent with the theoretical formula Var = Delta^2 / 12.\n');
fprintf('   - Simulated uniform and K-means variances match theoretical values closely.\n\n');

fprintf('3. SQNR Performance (Task 4):\n');
fprintf('   - SQNR increases by approximately 6 dB for each added bit of resolution\n');
fprintf('     (L = 2 -> ~6 dB, L = 4 -> ~12 dB, L = 8 -> ~18 dB).\n');
fprintf('   - Both techniques match the theoretical 6.02 dB/bit rule closely.\n\n');

fprintf('4. Uniform vs. K-means Quantization:\n');
fprintf('   - Uniform quantizer determines levels analytically prior to observing data.\n');
fprintf('   - K-means quantizer determines levels iteratively based on data distribution.\n');
fprintf('   - For a uniform distribution, K-means naturally converges to uniform levels.\n');
fprintf('   - For non-uniform distributions (e.g. Gaussian), K-means would place more levels\n');
fprintf('     in high-density regions, significantly outperforming uniform quantization.\n');
fprintf('================================================================================\n\n');

%% 4. Plotting Results & Saving Figures

% Figure 1: Mean Absolute Error (MAE) vs Levels
h1 = figure('Name', 'Task 2 - Mean Absolute Error vs Levels', 'NumberTitle', 'off');
plot(levels, mae_uniform, '-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'Uniform Quantizer');
hold on;
plot(levels, mae_kmeans, '--s', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'K-means Quantizer');
grid on;
```

```
xlabel('Number of Quantization Levels (L)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Mean Absolute Error (MAE)', 'FontSize', 11, 'FontWeight', 'bold');
title('Task 2: Mean Absolute Quantization Error vs Number of Levels', 'FontSize', 12, 'FontWeight', 'bold');
legend('Location', 'northeast', 'FontSize', 10);
xticks(levels);
saveas(h1, 'task2_mae_vs_levels.png');

% Figure 2: Quantization Error Variance vs Levels
h2 = figure('Name', 'Task 3 - Error Variance vs Levels', 'NumberTitle', 'off');
plot(levels, var_uniform, '-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'Uniform (Simulated)');
hold on;
plot(levels, var_kmeans, '--s', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'K-means (Simulated)');
plot(levels, var_theory, ':d', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'Theoretical (\Delta^2 / 12)');
grid on;
xlabel('Number of Quantization Levels (L)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Quantization Error Variance', 'FontSize', 11, 'FontWeight', 'bold');
title('Task 3: Quantization Error Variance vs Number of Levels', 'FontSize', 12, 'FontWeight', 'bold');
legend('Location', 'northeast', 'FontSize', 10);
xticks(levels);
saveas(h2, 'task3_variance_vs_levels.png');

% Figure 3: SQNR vs Levels
h3 = figure('Name', 'Task 4 - SQNR vs Levels', 'NumberTitle', 'off');
plot(levels, sqnr_uniform, '-o', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'Uniform (Simulated)');
hold on;
plot(levels, sqnr_kmeans, '--s', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'K-means (Simulated)');
plot(levels, sqnr_theory, ':d', 'LineWidth', 1.8, 'MarkerSize', 8, 'DisplayName', 'Theoretical (20 log_{10}(L))');
grid on;
xlabel('Number of Quantization Levels (L)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('SQNR (dB)', 'FontSize', 11, 'FontWeight', 'bold');
title('Task 4: Signal-to-Quantization-Noise Ratio (SQNR) vs Number of Levels', 'FontSize', 12, 'FontWeight', 'bold');
legend('Location', 'southeast', 'FontSize', 10);
xticks(levels);
saveas(h3, 'task4_sqnr_vs_levels.png');

fprintf('Simulation complete. Three figures displayed and saved as PNG images:\n');
fprintf('  - task2_mae_vs_levels.png\n');
fprintf('  - task3_variance_vs_levels.png\n');
fprintf('  - task4_sqnr_vs_levels.png\n');
```